

JEE Main - 5 | JEE - 2023 (Gen 1 & 2)

Date: 19/10/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

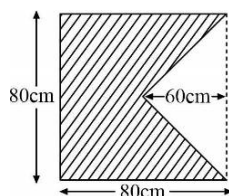
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

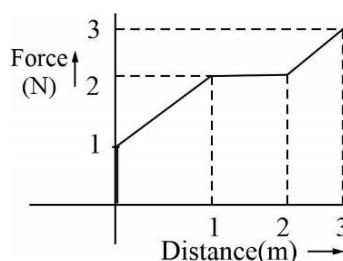
PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- A force acts on a 1 kg object so that its position is given as a function of time as $x = 4t^2$. What is the work done by this force in first 5 seconds?
(A) 950 J (B) 800 J (C) 875 J (D) 850 J
- Two masses of 4 kg and 9 kg are moving with equal kinetic energy. The ratio of magnitude of their linear momentum is:
(A) $\frac{4}{9}$ (B) $\frac{3}{2}$ (C) $\frac{9}{4}$ (D) $\frac{2}{3}$
- A spring of force constant 100 N/m has an extension of 1 cm. The work done in extending it from 1 cm to 5 cm is:
(A) 2 J (B) 1 J (C) 0 J (D) 0.12 J
- An isosceles triangular plane is symmetrically cut out of a square plate of side length 80 cm. The distance of centre of mass of the remaining plane with respect to centre of square will be:

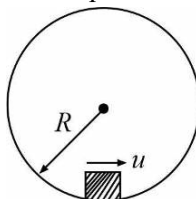


- (A) 24 cm (B) 10.9 cm (C) 12 cm (D) 5.45 cm
- A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is :

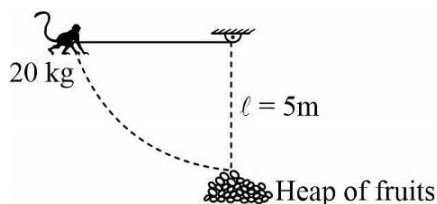


- (A) 3 J (B) 5 J (C) 6 J (D) 4 J
- A ball starts to fall from a height H on a floor with coefficient of restitution between floor and ball e . Find the height the ball goes up after second collision.
(A) $H \times e^2$ (B) $H \times e$ (C) $H \times e^4$ (D) $H \times 2e$
 - If the potential energy between two molecules is given by $U = \frac{-A}{r^7} + \frac{B}{r^{14}}$, then at equilibrium, separation between molecules, is :
(A) $(A/B)^{1/7}$ (B) $(2A/B)^{1/7}$ (C) $(2A/B)^{1/7}$ (D) $(2B/A)^{1/7}$

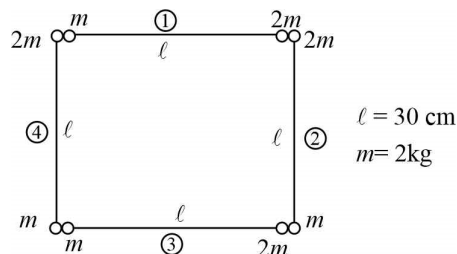
8. A particle is given an initial speed u inside a smooth spherical shell of radius $R = 1\text{m}$ such that it is just able to complete the circle. Acceleration of the particle when it is at the top point is :



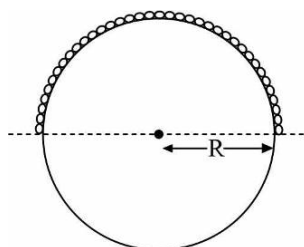
- (A) $g\sqrt{10}$ (B) g (C) $g\sqrt{2}$ (D) $g\sqrt{6}$
9. A monkey of mass 20 kg starts swinging on a light rope of length 5m from rest as shown in figure. Every time it passes above the heap of fruits it picks 1 kg fruits. The speed of monkey just after it passes fifth time above the heap will be: ($g = 10\text{m/s}^2$)



- (A) 5 m/s (B) 4 m/s (C) 8 m/s (D) 10 m/s
10. Four asymmetric dumbbells are arranged as shown above. If dumbbell (1) and dumbbell (3) are rotated by 180° then the COM of system shifts by a distance of:
- (A) 5 cm towards right
(B) 5 cm towards left
(C) 10 cm towards right
(D) 10 cm towards left

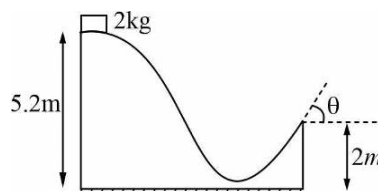


11. A chain of mass per unit length 0.2 kg/m is kept on a smooth horizontal cylinder having radius $R = \pi m$ as shown in figure. The chain is slightly disturbed as allowed to fall. Its speed as it leaves the cylinder will be: [Take $\pi^2 = 10$ and $g = 10\text{ m/s}^2$]

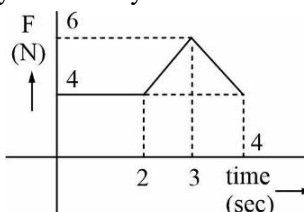


- (A) $\sqrt{120}\text{ m/s}$ (B) $\sqrt{130}\text{ m/s}$ (C) $\sqrt{140}\text{ m/s}$ (D) $\sqrt{150}\text{ m/s}$
12. A massless spring is tied to a ceiling. The spring constant of spring is 100N/m . If a block of mass 2kg is attached to the bottom of the spring in its original length and left, find the maximum elongation in the spring. {Take $g = 10\text{m/s}^2$ }
- (A) 0.3 m (B) 0.2 m (C) 0.4 m (D) 0.6 m

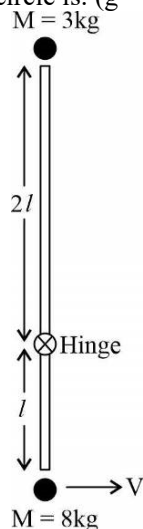
13. Two identical blocks are simultaneously released from the tops of two smooth tracks that have identical shapes (as shown in figure) but different angles of projection. If $\theta = 30^\circ$ for first track and $\theta = 45^\circ$ for second track then difference between maximum heights attained by blocks in air will be: ($g = 10 \text{ m/s}^2$)



- (A) 0.6 m (B) 0.4 m (C) 1 m (D) 0.8 m
14. A batsman hits a cricket ball coming towards him with a velocity of 15 m/s such that its velocity becomes 20 m/s in direction of bowler. If the mass of the ball is 200 grams , find the impulse imparted by the batsman to the ball.
- (A) 1 N-s (B) 3 N-s (C) 7 N-s (D) 2 N-s
15. A rod of length L is kept on x axis such that its one end is at origin and other is at $x = L$. The mass density of the rod varies as $\lambda(x) = \lambda_0 x^2$, where λ_0 is a constant and x is the distance from origin. Find the centre of mass coordinate of rod.
- (A) $L/4$ (B) L (C) $L/2$ (D) $3L/4$
16. A body of mass 2 kg moving with an initial velocity of 2 m/s is applied a time-varying force as shown in graph. Find the final kinetic energy of the body.

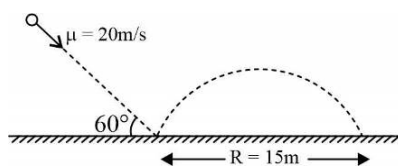


- (A) 121 J (B) 81 J (C) 242 J (D) 162 J
17. Two point masses of 8 kg and 3 kg are connected by a 12 m light rod. The rod is hinged at 4 m distance from 8 kg particle and it is free to rotate in vertical plane. The minimum speed that should be given to 8 kg particle so that it completes a vertical circle is: ($g = 10 \text{ m/s}^2$)

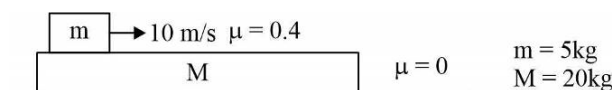


- (A) 6 m/s (B) 8 m/s (C) 4 m/s (D) 5.2 m/s

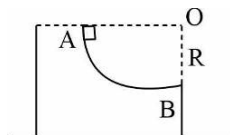
18. A particle strikes smooth floor with speed 20 m/s as shown in figure. After the collision it performs a projectile motion having range 15 m . The coefficient of restitution for the collision is: ($g = 10\text{ m/s}^2$)



- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{3}{4}$ (C) $\frac{2}{3}$ (D) $\frac{\sqrt{3}}{4}$
19. A 5 kg block is projected on the surface of a 20 kg plank with 10 m/s speed. The plank is sufficiently long and is initially at rest. Given that the floor below the plank is perfectly smooth, magnitude of work done by friction on the block will be: ($g = 10\text{ m/s}^2$)



- (A) 250 J (B) 240 J (C) 200 J (D) 10 J
20. A smooth Semicircular wedge of mass M is kept on a smooth floor and a small block of mass m is kept at its top point A as shown in figure. The block is let go and it slides till point B. Find the distance moved by the wedge when the block reaches point B.



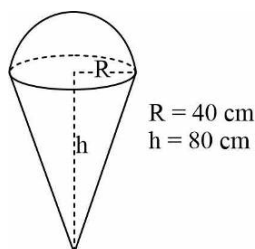
- (A) $\frac{MR}{M+m}$ (B) R (C) $\frac{mR}{m+M}$ (D) $\frac{R}{2}$

SPACE FOR ROUGH WORK

SECTION-2

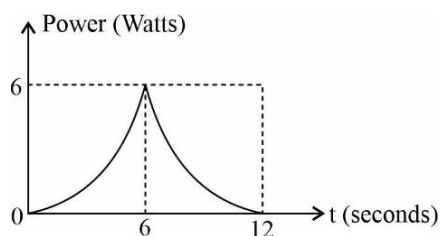
This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. A giant ice cream cone, as shown in figure, is completely filled with ice-cream.



Assuming both cone and ice cream are made up of material having same density, the distance of COM of the system from their common centre is $\frac{10}{\beta} \text{ cm}$. Then β is _____.

22. A particle of mass 6 kg receives power as per graph shown below. The power vs time graph consists of two circular arcs. If the initial velocity of particle is 2 m/s then its velocity at $t = 12$ sec. will be _____ m/s. (Take $\pi \times 6^2 = 114$)



23. A constant Power of 2W is being supplied to a block of mass 1kg kept at rest on ground. Find its velocity (in m/s) after 4 seconds.
24. A block of mass 2 kg is moving with a speed of 4 ms^{-1} on a smooth surface. It strikes another mass of 6 kg and then they move together as a single body. The kinetic energy loss (in Joules) during the collision is x . Find x .
25. A 60 kg bomb at rest explodes into three identical parts. If two parts start moving with 25 m/s speed each, and making 120° with each other, then the energy released during explosion is $(2^p)(3^q)(5)^r$ joules. The value of $p + q + r$ is _____.

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

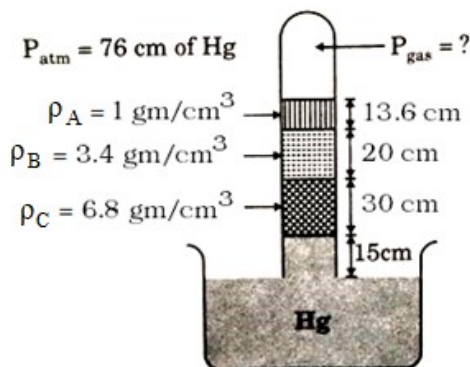
- What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure?
 (A) 80 cal (B) 540 cal (C) 620 cal (D) zero
- If two gases A and B and temperatures T_A and T_B respectively have identical Maxwellian plots, then which of the following statement is **true** ?
 (A) $T_B = T_A$
 (B) $M_B = M_A$
 (C) $\frac{T_A}{M_B} = \frac{T_B}{M_A}$
 (D) Gases A and B may be O_2 and SO_2 at 27°C and 327°C respectively
- Standard entropy of X_2 , Y_2 and XY_3 are 60, 40 and $50\text{JK}^{-1}\text{mol}^{-1}$, respectively. For the reaction,
 $\frac{1}{2}\text{X}_2 + \frac{3}{2}\text{Y}_2 \longrightarrow \text{XY}_3$, $\Delta H = -30\text{kJ}$, to be at equilibrium the temperature will be :
 (A) 500 K (B) 750 K (C) 1000 K (D) 1250 K
- Given the following equations and ΔH° values, determine the enthalpy of reaction at 298 K for the reaction :

$$\text{C}_2\text{H}_4(\text{g}) + 6\text{F}_2(\text{g}) \longrightarrow 2\text{CF}_4(\text{g}) + 4\text{HF}(\text{g})$$

$\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \longrightarrow 2\text{HF}(\text{g});$	$\Delta H_1^\circ = -537\text{kJ}$
$\text{C}(\text{s}) + 2\text{F}_2(\text{g}) \longrightarrow \text{CF}_4(\text{g});$	$\Delta H_2^\circ = -680\text{kJ}$
$2\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g});$	$\Delta H_3^\circ = -52\text{kJ}$

 (A) -1165 (B) -2382 (C) +1165 (D) +2382
- The temperature of 1 mole of an ideal gas is increased by 1°C at constant pressure. The work done is :
 [R is gas constant]
 (A) -R (B) -2R (C) $-\frac{R}{2}$ (D) -3R
- What will be the value of resonance energy of N_2O if given that:
 $\Delta H_{(\text{N}=\text{N})}^\circ = 400\text{kJ/mol};$
 $\Delta H_f^\circ (\text{N}_2\text{O}) = 100\text{kJmol}^{-1}$
 $\Delta H_{(\text{N}\equiv\text{N})}^\circ = 950\text{kJ/mol};$
 $\Delta H_{(\text{N}=\text{O})}^\circ = 600\text{kJ/mol}$
 $\Delta H_{(\text{O}=\text{O})}^\circ = 500\text{kJ/mol}$
 (A) -200 kJ/mol (B) -150 kJ/mol (C) -100 kJ/mol (D) -50 kJ/mol

7. In the figure shown the pressure of the confined gas will be : (A, B, C are immiscible liquids and also not soluble in liquid Hg)



- (A) 30 cm of Hg (B) 40 cm of Hg (C) 36 cm of Hg (D) 46 cm of Hg
8. Which of the following reactions is associated with the **most negative** change in entropy ?
 (A) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$ (B) $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g})$
 (C) $\text{C}(\text{s, graphite}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ (D) $3\text{C}_2\text{H}_2(\text{g}) \longrightarrow \text{C}_6\text{H}_6(\ell)$
9. Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. If enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol . Calculate $\Delta H_{\text{ionization}}$ of H_3PO_3 into its ions :
 (A) 50.84 kJ/mol (B) 5 kJ/mol (C) 2.5 kJ/mol (D) None of these
10. Certain volume of pure O_2 diffuses through an aperture in 120 seconds, whereas a same volume of mixture of O_2 and another gas, containing 60% O_2 diffuses from the same aperture in 180 seconds under similar conditions of pressure and temperatures, molecular weight of gas the other is :
 (A) 132 (B) 66 (C) 72 (D) 46.6
11. A sample of gas is compressed from an initial volume of $2V_0$ to V_0 using three different processes.
First : Using reversible isothermal
Second : Using reversible adiabatic
Third : Using irreversible adiabatic under a constant external pressure, then :
 (A) Final temperature of gas will be highest at the end of third process.
 (B) Final temperature of gas will be highest at the end of second process.
 (C) Enthalpy change of sample will be highest in first process. (magnitude wise)
 (D) Final temperature of gas will be highest at the end of first process.
12. The temperature of a certain mass of a gas increased from 37°C to 38°C . The volume of gas :
 (A) will decrease by $\frac{1}{273}$ of its volume at 37°C
 (B) will increase by $\frac{1}{273}$ of its volume at 273 K
 (C) will increase by $\frac{1}{273}$ of its volume at 310 K
 (D) will remain the same

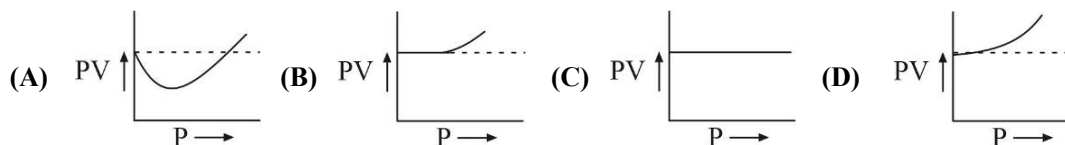
13. The temperature of a definite amount of an ideal monoatomic gas becomes four times in a reversible process for which heat exchange is zero. Which of the following is correct relation between the final and initial parameters of gas ?

(A) $V_f = 8V_i$ (B) $P_f = 32P_i$ (C) $V_f = 16V_i$ (D) $P_f = \frac{1}{16}P_i$

14. Enthalpy of polymerisation of ethylene, as represented by the reaction, $n\text{CH}_2 = \text{CH}_2 \longrightarrow (-\text{CH}_2 - \text{CH}_2 -)_n$ is -100kJ per mole of ethylene. Given bond enthalpy of $\text{C} = \text{C}$ bond is 600kJ mol^{-1} , enthalpy of $\text{C} - \text{C}$ bond (in kJ mol^{-1}) will be :

(A) 116.7 (B) 350 (C) 700 (D) Indeterminate

15. At critical temperature, which of the following graph is correct ?



16. The work done in adiabatic compression of 2 mole of an ideal monoatomic gas by constant external pressure of 2 atm starting from initial pressure of 1 atm and initial temperature of 300 K is:

($R = 2\text{ cal / mol }^\circ\text{C}$)

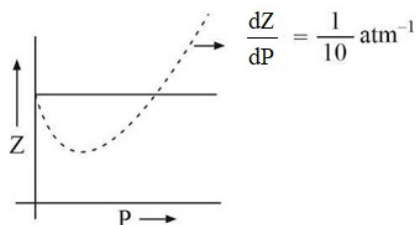
(A) 360 cal (B) 720 cal (C) 800 cal (D) 1000 cal

17. The graph of compressibility factor (Z) vs P for one mole of a real gas is shown in following diagram.

The graph is plotted at constant temperature 273 K. If the slope of graph at very high pressure $\left(\frac{dZ}{dP}\right)$ is

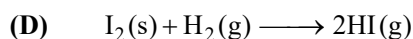
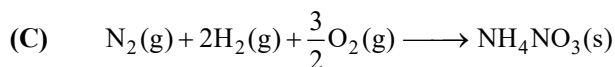
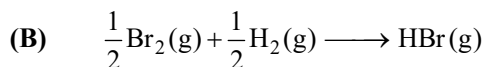
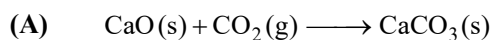
$\frac{1}{10}\text{ atm}^{-1}$, the volume of one molecule of real gas in cm^3 is :

[Given : $R = \frac{22.4}{273}\text{ L atm K}^{-1}\text{ mol}^{-1}$ and $N_A = 6 \times 10^{23}$]



(A) 9.3×10^{-23} (B) 3.7×10^{-20} (C) 9.3×10^{-22} (D) 5.6×10^{-20}

18. Which of the reaction defines molar enthalpy of formation of the respected products?



19. Which of the following is incorrect expression of rate of diffusion/effusion. At constant temperature & pressure.

$$\left[\begin{array}{l} V \rightarrow \text{Volume of gas diffused} \\ n \rightarrow \text{Number of moles of gas diffused} \\ \text{Where } w \rightarrow \text{Mass of gas diffused in} \\ d \rightarrow \text{Distance travelled by gas molecules} \\ M \rightarrow \text{Molar mass of the gas diffused} \end{array} \right]$$

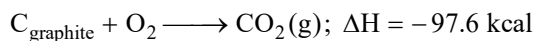
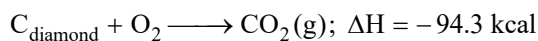
(A) $\frac{V_1}{V_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$

(B) $\frac{w_1}{w_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_1}{M_2}}$

(C) $\frac{n_1}{n_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_1}{M_2}}$

(D) $\frac{d_1}{d_2} \times \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$

20. For the following reaction,



The heat required to convert 1g of diamond completely into graphite is :

- (A) 1.59 kcal (B) 0.1375 kcal (C) 0.55 kcal (D) 0.275 kcal

SPACE FOR ROUGH WORK

SECTION-2

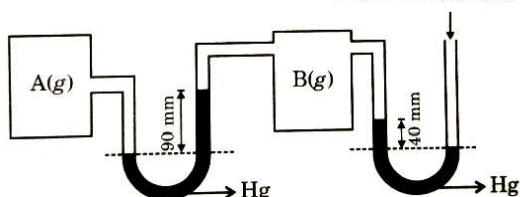
This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. A sample of argon gas of mass 12.0 g is expanded reversibly adiabatically from 1.0 L at 294 K to 3.0 L.

The value of final temperature is 'T'. What is the value of $\frac{T}{28}$? (Take: $3^{2/3} = 2.1$)

22. At 300 K, two gases are filled in two equal sized containers as given.

$$1 \text{ atm} = 760 \text{ mm of Hg}$$



The pressure of gas A is P (cm of Hg). What is the value of $\frac{P}{27}$?

23. A gas expands from a volume of 3 dm^3 to 30 dm^3 against a constant pressure of 7 bar initially at 27°C . If heat energy equal to magnitude of the work done during expansion is used to heat 50 moles of water then rise in temperature (K) of water is: (Take specific heat capacity of H_2O as $4.2 \text{ J/grams} \cdot \text{K}$ and $1 \text{ litre bar} = 100 \text{ J}$)
24. Two gases A and B having molecular weights 60 and 45 respectively are enclosed in a vessel. The weight of A is 0.5 g and that of B is 0.2 g. The total pressure of the mixture is 750 mm. Partial pressure (in mm) of gas B is 'Y'. What is the value of $\frac{Y}{52}$?
25. Molar enthalpy of vaporisation of water at 1 atm, 300 K is 1100 cal/mole. If 2 mole of water is vaporised in a closed container at 300 K then change in internal energy is $x \times 10^3 \text{ cal}$. What is the value of x. $[R = 2 \text{ cal/mole} \cdot \text{K}]$ (Assume ideal behaviour of $\text{H}_2\text{O(g)}$)

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A , B and C are placed at the points $(1, 2)$, $(2, 2)$ and $(4, 4)$ respectively. Then which of these stones is/are on the path of the man?
(A) B only (B) A only (C) All the three (D) C only
- The image of the point $(3, 5)$ in the line $x - y + 1 = 0$, lies on :
(A) $(x - 2)^2 + (y - 4)^2 = 4$ (B) $(x - 4)^2 + (y + 2)^2 = 16$
(C) $(x - 4)^2 + (y - 4)^2 = 4$ (D) $(x - 2)^2 + (y - 2)^2 = 4$
- The intersection of three lines $x - y = 0$, $x + 2y = 3$ and $2x + y = 6$ is a:
(A) Equilateral triangle (B) Right angled triangle
(C) Isosceles triangle (D) None of these
- The line $2x - y + 1 = 0$ is a tangent to the circle at the point $(2, 5)$ and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is:
(A) $3\sqrt{5}$ (B) $5\sqrt{3}$ (C) $5\sqrt{4}$ (D) $4\sqrt{5}$
- Choose the correct statement about two circles whose equations are given below:
$$x^2 + y^2 - 10x - 10y + 41 = 0$$
$$x^2 + y^2 - 22x - 10y + 137 = 0$$

(A) Circles have same centre (B) Circles have no meeting point
(C) Circles have only one meeting point (D) Circles have two meeting points
- A ray of light coming from the point $(1, 2)$ is reflected at a point A on the x -axis and then passes through the point $(5, 3)$. The coordinates of the point A are :
(A) $(13/5, 0)$ (B) $(12/5, 0)$ (C) $(3, 0)$ (D) None of these
- If the co-ordinates of the middle point of the portion of a line intercepted between coordinate axes is $(3, 2)$, then the equation of the line will be :
(A) $2x + 3y = 12$ (B) $3x + 2y = 12$ (C) $4x - 3y = 6$ (D) $5x - 2y = 10$
- The equation of perpendicular bisectors of the sides AB and AC of a triangle ABC are $x - y + 5 = 0$ and $x + 2y = 0$ respectively. If the point A is $(1, -2)$, then the equation of line BC is :
(A) $23x + 14y - 40 = 0$ (B) $14x - 23y + 40 = 0$
(C) $14x - 23y + 236 = 0$ (D) $14x + 23y - 40 = 0$

9. The medians AD and BE of a triangle with vertices $A(0, b)$, $B(0, 0)$ and $C(a, 0)$ are perpendicular to each other, if :
- (A) $a = \sqrt{2}b$ (B) $a = -\sqrt{2}b$
 (C) Both (A) and (B) (D) None of these
10. Given vertices $A(1, 1)$, $B(4, -2)$ and $C(5, 5)$ of a triangle, then the equation of the perpendicular dropped from C to the internal bisector of the angle A is :
- (A) $y - 5 = 0$ (B) $x - 5 = 0$ (C) $x + 5 = 0$ (D) $2x + 5 = 0$
11. A line L passes through the points $(1, 1)$ and $(2, 0)$ and another line L' passes through $\left(\frac{1}{2}, 0\right)$ and perpendicular to L . Then the area of the triangle formed by the line L , L' and y -axis, is:
- (A) $15/8$ (B) $25/5$ (C) $25/8$ (D) $25/16$
12. If the length of tangent drawn from the point $(5, 3)$ to the circle $x^2 + y^2 + 2x + ky + 17 = 0$ be 7, then $k =$
- (A) 4 (B) -4 (C) -6 (D) None of these
13. The line $lx + my + n = 0$ will be a tangent to the circle $x^2 + y^2 = a^2$ if :
- (A) $n^2(l^2 + m^2) = a^2$ (B) $a^2(l^2 + m^2) = n^2$
 (C) $nl = am$ (D) $a(l + m) = n$
14. If OA and OB be the tangents to the circle $x^2 + y^2 - 6x - 8y + 21 = 0$ drawn from the origin O , then $AB =$
- (A) 11 (B) $\frac{4}{5}\sqrt{21}$ (C) $\sqrt{\frac{17}{3}}$ (D) None of these
15. If the line $y = mx + c$ be a tangent to the circle $x^2 + y^2 = a^2$, then the point of contact is:
- (A) $\left(\frac{-a^2}{c}, a^2\right)$ (B) $\left(\frac{a^2}{c}, \frac{-a^2m}{c}\right)$ (C) $\left(\frac{-a^2m}{c}, \frac{a^2}{c}\right)$ (D) $\left(\frac{-a^2c}{m}, \frac{a^2}{m}\right)$
16. In a $\triangle ABC$, side AB has the equation $2x + 3y = 29$ and the side AC has the equation $x + 2y = 16$. If the mid point of BC is $(5, 6)$, then the equation of BC is :
- (A) $2x + y = 16$ (B) $x + y = 11$ (C) $2x - y = 4$ (D) $x - y + 1 = 0$
17. If the equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of c is :
- (A) $\frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$ (B) $a_1^2 - a_2^2 + b_1^2 - b_2^2$
 (C) $\frac{1}{2}(a_1^2 + a_2^2 + b_1^2 + b_2^2)$ (D) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$
18. The equation of angle bisectors of two lines L_1 and L_2 are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line L_1 passes through $(-11, 4)$, the equation of acute angle bisector of L_1 and L_2 is :
- (A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$
 (C) $2x + 16y + 5 = 0$ (D) None of these

19. If $P\left(1 + \frac{t}{\sqrt{2}}, 2 + \frac{t}{\sqrt{2}}\right)$ be any point on a line, then the complete range of values of t for which the point P lies between the parallel lines $x + 2y = 1$ and $2x + 4y = 15$ is :
- (A) $-\frac{4\sqrt{2}}{3} < t < \frac{5\sqrt{2}}{6}$ (B) $0 < t < \frac{5\sqrt{2}}{6}$ (C) $-\frac{4\sqrt{2}}{5} < t < 0$ (D) $-\frac{4\sqrt{2}}{3} < t < \frac{\sqrt{2}}{6}$
20. The normal at the point $(3, 4)$ on a circle cuts the circle at the point $(-1, -2)$. Then the equation of the circle is :
- (A) $x^2 + y^2 + 2x - 2y - 13 = 0$ (B) $x^2 + y^2 - 2x - 2y - 11 = 0$
(C) $x^2 + y^2 - 2x + 2y + 12 = 0$ (D) $x^2 + y^2 - 2x - 2y + 14 = 0$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. Let a point P be such that its distance from the point $(5, 0)$ is thrice the distance of P from the point $(-5, 0)$. If the locus of the P is a circle of radius r , then $4r^2$ is equal to $\frac{p}{q}$ (p, q are co-prime) then $p - 55q$ is _____.
22. If the locus of the mid-point of the line segment from the point $(3, 2)$ to a point on the circle $x^2 + y^2 = 1$ is a circle of the radius r , then $10r$ is equal to :
23. Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$ ($a < 0$) be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to $\sqrt{n}$. The value of n is _____.
24. The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$x^2 + y^2 - 24x - 10y + 160 = 0$$
 is _____.
25. If $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$ and $\left(d, \frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units, then $abcd$ is equal to :

SPACE FOR ROUGH WORK

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